

Vocabulary resonance in US trademark filings: Cross-industry diffusion structure and the U-shape in registration outcomes

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Abstract

The DeDeo prospective/retrospective Kullback–Leibler resonance framework (Barron et al., 2018; Murdock et al., 2017) was developed on cognitive corpora in which the author of each document is the agent whose novelty is being measured: Darwin’s reading notebooks, parliamentary speeches of the French Revolution. This paper asks whether the framework produces interpretable structure on a qualitatively different corpus type — the goods/services descriptions of US trademark filings, which are commercially-incentivised text drafted by counsel to define legal protection scope. The corpus is 2.32 million granted USPTO filings, 1990–2024, across four NICE classes selected to exercise a canonical cross-industry phenomenon (software 009, tech-services 042, transport 039, advertising/retail 035).

The construct, denoted ΔKL , produces three empirical patterns. **First**, cross-industry vocabulary diffusion is directly traceable. Software- and tech-services-origin themes (cloud, AI, mobile-app, as-a-service, blockchain, streaming, AR/VR) arrive in transport and advertising/retail with multi-year lags of 1–13 years. At the theme level via LDA topic extraction, the same asymmetric flow reproduces: of 33 themes reaching at least two classes, 33 outflow edges originate in software/tech-services and only 4 return there. Bass diffusion fits on 118 (theme, class) trajectories with $R^2 \geq 0.7$ give median $q/p \approx 7$, in the range typically reported for consumer-product adoption. **Second**, ΔKL produces a U-shape in post-registration §8 maintenance outcomes. At the filing level, filings whose vocabulary moves with the field and filings whose vocabulary moves against it both maintain registration 5–10 percentage points more often than the flux-neutral middle. At the token level the gap is sharper: marks containing distinctive-tail tokens (typesetting, dial-up, blockchain, NFT) survive at 50–80% while marks containing flux-neutral middle tokens (organisational, conformity, freighting) survive at 11–26%. **Third**, the signal is orthogonal-to-negative with patents within firm: -0.048σ , $t = -7.9$ on 14,463 trademark-and-patent-matched firms. ΔKL is empirically distinct from patent-track invention.

A worked discipline result: the raw Schumpeter Mark I read of new themes (73.7% entrant-dominated) collapses on year-matched baseline contrast to a -2.7pp mean excess once the corpus’s 76.4% overall debut rate is netted out. The paper reports this rather than burying it. The paper defends a bounded reading: ΔKL is a measurement of lexical resonance with interpretable empirical structure on commercial text, not an innovation measure in the strong sense. Code and per-firm-year panels are released for replication at <https://github.com/ericssilver/novelty>.

1 Introduction

Murdock et al. (2017) introduced a measurement framework for directional novelty in time-ordered text corpora: for each document, compare its token distribution to aggregated distributions of

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past documents (prospective surprise) and future documents (retrospective surprise). Their signed difference — positive when a document is unusual against its recent past and close to its near future — they call *resonance*. The framework was developed on Charles Darwin’s reading notebooks and extended in Barron et al. (2018) to French Revolution parliamentary debates. Both corpora share a salient property: the text is authored by the agent whose novelty is the object of measurement, and the author’s cognitive or rhetorical state is what the framework is designed to track.

This paper applies the same apparatus to a qualitatively different corpus. The goods/services descriptions of US trademark filings are *commercially-incentivised*, typically drafted by counsel rather than the firm principal, and exist to define legal protection scope — not to express cognition or rhetoric. The text is subject to two competing pressures: it must be wide enough to cover the applicant’s intended commercial activity and narrow enough to be granted by an examiner. Whether the resonance framework produces interpretable structure on text of this kind is the empirical question of this paper.

The empirical record in text-based economics has converged on text-similarity methods — the text-based network industry classification of Hoberg & Phillips (2016) being the most influential — and has rarely used trademark text for measurement. Trademark filings have been used to study business formation (Dinlersoz et al., 2018) and to distinguish protective filings from substantive product introductions (Graham et al., 2013), but the text of those filings has been largely unused.

The paper’s claims are bounded by what the data support. The constructed signal ΔKL , computed as in Barron et al. (2018); Murdock et al. (2017), produces three empirical patterns on a 2.32 million-filing granted corpus 1990–2024 across four NICE classes. Cross-industry vocabulary diffusion is directly traceable at both the phrase and theme levels, with directional asymmetry: software- and tech-services-origin themes arrive in transport and advertising/retail years later, not the reverse. Bass-fittable adoption curves with median $q/p \approx 7$ on 118 (theme, class) trajectories lend external plausibility to the diffusion reading. The asymmetric flow reproduces under independently-fitted non-negative matrix factorization on the same vocabulary as a within-paradigm robustness check. Within firms, ΔKL is orthogonal-to-negative with patent counts, establishing the construct as empirically distinct from patent-track invention. At the filing-and-token level, ΔKL predicts post-registration §8 maintenance in a U-shape that is robust to operationalization but whose mechanism is not separable with the trademark record alone.

What the paper does not claim. ΔKL does not measure innovation in any strong sense. The within-firm patent-orthogonality refutes the “ ΔKL replaces patents as an innovation measure” reading; the framework’s status as commercially-incentivised text refutes the cognitive-state reading inherited from Darwin’s notebooks; and the small- n validation against expert “most innovative” lists ($n = 48$ matched firms) is too weak to license the strong claim. What ΔKL measures, on this evidence, is *lexical resonance* on commercial text: a filing’s forward-leaning position in its industry’s vocabulary trajectory, with interpretable empirical consequences that are documented and bounded below.

A worked methodological discipline is folded into Section 7: the Schumpeter Mark I read of new themes (entrant-led innovation) collapses on year-matched baseline contrast. The methodological point is that the trademark-text literature is particularly exposed to base-rate bias because the corpus is mechanically saturated in entrants, and the discipline of year-matched contrast is what distinguishes useful from inflated findings.

Section 2 describes the construct, the corpus, and known fragilities. Section 3 presents the cross-industry phrase transit table. Section 4 reports the U-shape in registration outcomes at the filing and token levels. Section 5 reports the theme-level evidence: Bass-fittable adoption curves and cross-class flow asymmetry under LDA topic extraction, with an NMF robustness check. Section 6 reports the within-firm patent orthogonality. Section 7 reports the Schumpeter Mark I baseline

collapse. Section 8 states scope. Section 9 is reproducibility.

2 Δ KL on a USPTO trademark corpus

2.1 The DeDeo apparatus

For a time-ordered sequence of documents, with each document at year t producing a token distribution P_t , the DeDeo framework defines two quantities against a window W of past and future documents (Barron et al., 2018; Murdock et al., 2017):

- *Prospective surprise*: $\text{KL}(P_t \parallel Q_{\text{past}})$, where Q_{past} is the aggregate token distribution of documents in years $t - W$ to $t - 1$.
- *Retrospective surprise*: $\text{KL}(P_t \parallel Q_{\text{future}})$, where Q_{future} is the aggregate distribution of documents in years $t + 1$ to $t + W$.

The signed resonance is

$$\Delta\text{KL}_t = \text{KL}(P_t \parallel Q_{\text{past}}) - \text{KL}(P_t \parallel Q_{\text{future}}).$$

Positive ΔKL marks a document whose vocabulary is unusual relative to recent past and close to near future. Negative ΔKL marks the inverse, language that looked normal at filing and looked played-out in hindsight.

2.2 Corpus

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. For this paper I work with four NICE classes selected to exercise canonical cross-industry diffusion: software (009), scientific and technical services (042), transport (039), and advertising and retail (035). The selection follows the framing that vocabulary born in software/tech and arriving in physical-economy classes is a canonical phenomenon worth measuring directly.

Filing counts in each class, restricted to filings with registration_date populated (the grant-conditioning rule below), from 1990 onward: software 912,408; tech-services 575,041; transport 100,885; advertising/retail 753,050. Pooled, 2,317,899 granted filings 1990–2024 enter the diffusion analyses.

2.3 Grant conditioning

The corpus is restricted to filings with a populated registration date. The conditioning has two motivations. First, applications include speculative or defensive filings that may not reflect realized commercial activity; the registration outcome screens out applications that did not survive examination. Second, the resonance signal is most interpretable for filings whose vocabulary reflects intended commercial use rather than maximal claim-scope drafting. Approximately 56% of Class 009 filings in the full backfile are granted; the four-class pooled share is similar. Grant conditioning also discards a substantial fraction of negative- ΔKL filings that may merely be unconcluded applications.

2.4 Computation

For each granted filing at year t in class c with goods/services text producing token distribution P (unigram + bigram, after a standard analyzer that drops stopwords and tokenizes within punctuation chunks), the prospective and retrospective reference distributions Q_{past} and Q_{future} are computed from all granted filings in class c in the respective windows. I use $W = 5$ years (flat window) with Dirichlet smoothing ($\alpha = 1$ per cell) to avoid infinite surprise from zero-count tokens. The within-class vocabulary is the set of unigrams and bigrams with document frequency ≥ 50 .

2.5 What the construct does and does not measure

The construct’s known fragilities have been documented in companion analysis on a complete Class 009 corpus of 1,807,636 filings. Prospective and retrospective KL are correlated at $r = +0.971$ across 1.56M scored filings; signed ΔKL is therefore a small residual of two large, highly correlated quantities. *This residual is noisy at the per-filing level.* The magnitude $|\Delta\text{KL}|$ inflates monotonically for short filings (0.293 for documents under 5 terms, decaying to 0.147 for documents of 40 or more terms) — a length artifact that requires control in any per-filing analysis. Pre-1995 readings are contaminated by thin past (mean ΔKL of +2.05 in 1990); the burn-in cut should exclude these years.

Window sensitivity is bounded. Across $W \in \{3, 5, 7\}$ years, only 6.9–10.4% of filings flip sign — the direction of the residual is robust to half-life kernel choice, the magnitude is kernel-relative. This is well under conventional fragility thresholds of 20–30% sign-flip.

What the construct measures, on this evidence, is *lexical resonance*: a filing’s forward-leaning position in its industry’s vocabulary trajectory at filing time. The construct does not measure innovation in the strong sense. Sections 3–6 examine what the construct produces empirically and what those patterns license.

3 Cross-industry phrase transit

3.1 Operationalization

To make cross-industry diffusion operational without committing to a specific topic-extraction method (Section 5 takes up the topic-extraction question), I define a list of 14 multi-token business-model phrases and track their prevalence across (class, year) cells directly. Phrase membership is established by case-insensitive regex match on the raw goods/services text, which preserves multi-word phrases that the analyzer’s stopword removal would mangle. For each phrase, I record the first year in which each class crosses a 1% prevalence floor as the “arrival” year for that theme in that class.

3.2 The transit table

Of the phrases that clear the 1% floor in transport and advertising/retail, almost all originate in software (009) or tech-services (042) and arrive in the physical-economy classes with lags of one to thirteen years. The exception is on-demand, which originates in transport (2014) before software/tech (2016–17). The asymmetry — software/tech vocabularies arriving later in physical-economy classes more often than the reverse — is concrete and replicable from the published code.

Table 1: First year a theme reaches 1% prevalence among granted filings, by class. “–” = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	2009	2013	2012
mobile-app	2010	2009	2012	2012
cloud	2012	2010	2011	2014
streaming	2008	2008	2013	2014
artificial-intelligence	2017	2016	2018	2018
augmented/virtual-reality	2015	2016	2022	2022
blockchain / crypto	2021	2018	–	2022
real-time	2001	2003	2009	2014
on-demand	2017	2016	2014	2017

3.3 Caveats

The 1% floor is sensitive to class size. Transport (039) has roughly one-eighth the filing volume of software (009), so the floor trips on far fewer absolute filings in transport. A phrase below 1% in transport may still be present with substantive absolute volume. The 1% choice is conventional; a different floor (0.5% or 2%) shifts arrival years by a year or two but does not reverse the directional pattern.

The asymmetry is also a four-class result. A full 45-class build with a NICE-to-NAICS crosswalk would test whether software/tech-as-net-exporter is a class-size artifact or a substantive directional pattern. The current evidence is consistent with the directional reading but cannot exclude a corpus-construction alternative.

4 The U-shape in post-registration outcomes

4.1 Filing-level result

Among granted filings in the post-burn-in 1996–2020 window with ≥ 10 terms, 5-year post-registration §8 maintenance is U-shaped in ΔKL . Filings whose ΔKL sits in the extreme positive tail and filings whose ΔKL sits in the extreme negative tail both maintain registration 5–10 percentage points more often than filings whose ΔKL sits near zero. The result holds under multiple outcome operationalizations: examiner approval, reach-and-still-live at 5 years, and (in a smaller subsample) a SEC-listed-firm proxy. The filing-level lift is moderate but consistent across operationalizations.

4.2 Token-level result

The U-shape is sharper at the token level. *Marks containing tokens at the extreme tails of within-class vocabulary trajectory survive substantially better than marks containing flux-neutral middle tokens.* Examples:

- Tokens at the negative tail — vocabulary that was disappearing from the class as the field moved on (*typesetting*, *dial-up*, *e-liquid*, *pre-craft beer*) — appear in marks that survive at 50–80%.
- Tokens at the positive tail — vocabulary that was emerging (*cannabis*, *crypto*, *vape*, *NFT*, several AI-era tokens) — appear in marks that survive at 50–82%.

- Tokens in the flux-neutral middle (*organisational, conformity, freighting, various*) appear in marks that survive at 11–26%.

The token-level gap is 30–50 percentage points. The filing-level statistic and the token-level statistic are different measurements: the filing-level statistic bins filings by their ΔKL score; the token-level statistic identifies tokens at the resonance tails of within-class vocabulary trajectories and reports survival of the marks that contain them. Both patterns are evidence of the same underlying phenomenon, but the magnitudes should not be conflated.

4.3 Robustness and mechanism

The U-shape is robust to substitution of the outcome operationalization (examiner approval, §8 maintenance at 5 years, currently-live at 10 years), to choice of half-life kernel within $W \in \{3, 5, 7\}$ years, to length control, and to class. The pattern is present in 32 of the 44 NICE classes that have sufficient post-1995 volume to estimate.

Mechanism is the harder question. Three candidate stories cannot be separated with the trademark record alone:

- *Intelligent harvesting* at the negative tail: applicants deliberately register sunseting categories when competition has departed, maintain the marks for residual revenue, and survive because the marks face little contest.
- *Passive late-occupation*: applicants register category-typical language inherited from before the field moved on, are not contested for the same reason, and survive by inertia rather than design.
- *Selection on something unobservable*: the kinds of applicants who file at the tails differ on capitalisation, sophistication, or other covariates that independently predict mark maintenance.

The trademark record does not separately identify which of these is the dominant mechanism. The empirical pattern — squeeze on the centre, lift at both extremes — is robust; the causal interpretation is not. Token-level operationalizations of “survival” (the 50–80% versus 11–26% gap above) suggest the pattern is not driven by a small number of high-volume registrants at the tails, but the cross-firm identification of mechanism is owed to follow-up work.

5 Theme-level evidence: Bass-fittable curves and cross-class flow asymmetry

The phrase transit in Section 3 depends on a curated phrase list. A theme-level analysis using unsupervised topic extraction provides independent corroboration: if the asymmetric flow is genuinely a corpus property rather than an artifact of which phrases I happened to pick, the same directional pattern should also appear at the theme level via topic-prevalence trajectories.

5.1 LDA topic extraction

I fit a Latent Dirichlet Allocation topic model with $T = 50$ topics on a stratified sample of 200,000 granted filings (50,000 per class, capped at class size; transport is smaller after sampling). The vocabulary is unigrams + bigrams with document frequency ≥ 200 , giving $V = 69,361$ terms. The fitted model is transformed across the full 2.32 million-filing pool to assign each filing a topic distribution and a top topic.

The resulting top-word lists are recognizable business-model components. Selected examples: T0 {health, care, wellness, healthcare}; T1 {autonomous, loading, unloading, truck, material handling}; T2 {store, retail, wholesale}; T10 {electronic transmission, computer networks, data, communication}; T30 {online, website, non-downloadable, information}; T49 {vehicles, traffic, bicycle, lights}. I make no claim that 50 unsupervised topics constitute the final theme set; cross-method stability and human face-validation remain owed before any individual theme is treated as headline.

5.2 Bass-fittable adoption curves

For each (theme, class) cell where a theme reaches the 1% prevalence floor and is tracked for at least six years post-arrival, I fit the Bass (1969) diffusion model to the normalized cumulative prevalence. The Bass model parameterizes adoption with an innovation coefficient p , an imitation coefficient q , and a saturation level m (Bass, 1969). I fit p and q with non-linear least squares, retaining fits with $R^2 \geq 0.7$.

Of 118 retained fits, the median p is 0.017, the median q is 0.114, and the median q/p is approximately 7. The median q/p falls in the range typically reported for consumer-product Bass fits in the diffusion-of-innovations literature, lending external plausibility to the (theme, class) adoption curves as describing real adoption events rather than topic-model artifacts. The bulk of fits (88 of 118) are platform-like (broad gradual adoption, moderate word-of-mouth); a small tail of 9 fits shows fad-shaped curves (high q/p).

5.3 Cross-class flow asymmetry, quantified

Using the 50 LDA themes, I count themes that reach at least the 1% floor in at least two classes and identify the origin (earliest class to cross the floor) and the second class (next class to cross). This produces a directed graph of theme flow between the four classes.

Table 2: Theme flow origin $\rightarrow$ second-arrival class.

Origin $\rightarrow$ second class	themes
software $\rightarrow$ tech-svcs	14
software $\rightarrow$ transport	11
software $\rightarrow$ advert/retail	8
transport $\rightarrow$ tech-svcs	7
transport $\rightarrow$ advert/retail	6
transport $\rightarrow$ software	5
tech-svcs $\rightarrow$ software	4
advert/retail $\rightarrow$ tech-svcs	5

Of 33 themes reaching at least two classes, 33 outflow edges originate in software (009 and 042 combined) and only 4 return there. Software and tech-services are the net exporters; the return flow is small. The asymmetry is at the theme level what the phrase-level transit table of Section 3 shows at the phrase level. The independent reproduction of the asymmetry under unsupervised topic extraction is the substantive corroboration of the diffusion claim.

5.4 Robustness under independently-fitted NMF

I refit topic structure independently using non-negative matrix factorization on the same 200,000-filing sample and vocabulary, with $T = 50$, then transform across the full pool. The Bass-fit pattern

and cross-class flow asymmetry reproduce. Seven of ten hand-curated seed phrases from Section 3 (blockchain, cloud, AI/ML, SaaS, mobile-app, AR/VR, marketplace) have a clear best-NMF-topic; the three missing (streaming, IoT, on-demand) are either too generic at the topic level or too brief at the phrase level to concentrate in a single topic at $T = 50$. The substantive diffusion findings do not depend on the choice between LDA and NMF, though both are bag-of-words matrix factorizations on the same vocabulary, so this robustness is within-paradigm; a cross-paradigm test under a sentence-embedding cluster or graph-based community detection method on the same corpus is the natural methodological extension and is not run here.

6 Empirical distinctness from patents

If ΔKL and patent counts were noisy measures of the same underlying construct, they should correlate within firms over time. They do not.

The within-firm analysis uses a published panel of 174,569 firm-years covering 14,463 trademark owners that have been name-matched to PatentsView patent assignees (PatentsView, 2024). The panel is constructed in companion work and used here only to validate the construct against the patent-track measure. I fit a two-way (firm + year) fixed-effects regression of standardized firm-year mean ΔKL on $\log(1 + \text{patents})$ with firm-clustered standard errors. The standard deviation of mean ΔKL on the collapsed firm-year panel is 0.214.

Table 3: Within-firm ΔKL versus patenting. Coefficients in standard deviations of ΔKL per unit $\log(1 + \text{patents})$.

Specification	coef (σ)	SE	t
Between-firm correlation (firm means)	−0.020	—	—
Within-firm correlation (firm+year demeaned)	−0.015	—	—
Two-way FE, contemporaneous	−0.048	0.006	−7.9
robustness: patents = max over matched assignees	−0.048	0.006	−8.0
controlling $\log(n_{\text{filings}})$	−0.051	0.006	−8.4
Lead/lag: patents $_{t-1}$	−0.054	0.009	−5.8
patents $_t$	−0.019	0.010	−1.9
patents $_{t+1}$	+0.010	0.010	+1.0

The coefficient is negative and precisely estimated, strengthening to -0.051 under filing-count control. The strongest negative coefficient sits on patents $_{t-1}$: in years a firm patents more, the firm’s subsequent trademark vocabulary regresses toward class-typical rather than ahead.

The within-firm result establishes that ΔKL is empirically distinct from patent-track invention. It does not by itself license a strong claim that ΔKL measures a different innovation channel. An alternative reading consistent with the within-firm data is that firm activity allocation between patent prosecution and brand work shifts year-to-year and the trademark vocabulary becomes more boilerplate during years of high patent output. The empirical distinctness is robust; the substantive interpretation of the distinct channel is bounded by the validation evidence elsewhere.

7 A worked discipline result: Schumpeter Mark I collapses on baseline contrast

This section reports a methodological result more than a substantive one, because the trademark-text literature is particularly exposed to base-rate bias on questions where raw shares are inter-

pretable as evidence.

The classical Schumpeter Mark I / II debate asks whether new innovations are predominantly born from new entrants (Mark I) or from incumbent diversification (Mark II) (Schumpeter, 1942). The standard empirical form is to identify a population of early adopters of an innovation, classify them as entrant or incumbent, and report the entrant share.

I apply this standard form to the 50 LDA themes. For each theme, I take its 50 earliest carriers (sorted by filing year, conditional on top-topic weight ≥ 0.20). I classify each carrier’s owner as an entrant if its first granted mark across the four-class corpus is in the carrier-filing year or one year earlier — that is, the carrier is filing at or near the owner’s debut.

Table 4: Raw entrant share among the 50 earliest carriers per theme.

Statistic	Value
Themes scored	50
Mean entrant share among 50 earliest carriers	73.7%
Themes entrant-dominated ($\geq 50\%$ entrants)	48 / 50
Themes incumbent-dominated ($\leq 20\%$ entrants)	0 / 50

The raw numbers look like strong Schumpeter Mark I. This is the result that does not survive baseline contrast.

The corpus’s overall debut rate among granted filings in 1990–2024 is 68–77% per year across the four classes. With a year-weighted mean (weighting by each theme’s earliest-carrier year distribution), the baseline is 76.4%. *Most filings in the four-class corpus over this period are by owners debuting that year.*

Table 5: Schumpeter I with year-matched population baseline.

Statistic	Value
Theme entrant share (mean / median)	73.7% / 73.0%
Year-matched baseline (mean / median)	76.4% / 76.7%
Excess (theme – baseline)	–2.7pp / –3.7pp
Themes with positive excess	22 / 50
Themes with excess $> 10\text{pp}$	9 / 50
Themes with negative excess	28 / 50

The mean theme excess is -2.7 percentage points; 28 of 50 themes are *below* baseline. Only nine themes show real entrant excess above 10pp. The Schumpeter Mark I claim collapses: themes inherit the corpus’s high debut rate without adding to it.

The methodological point: the trademark-text literature is particularly exposed to base-rate bias because the corpus is mechanically saturated in entrants, and any analysis that reports raw population shares without year-matched baseline contrast risks reading mechanical corpus structure as substantive finding. I report this result not because Schumpeter Mark I is the central question of the paper but because the discipline of base-rate contrast is what distinguishes useful from inflated findings in this kind of analysis, and the discipline should be visible in any work that uses trademark text to ask population-share questions.

8 Scope and limits

The paper makes the following claims.

Δ KL, computed as in Barron et al. (2018); Murdock et al. (2017) with a 5-year flat window on granted USPTO trademark filings, produces a directly traceable cross-industry vocabulary diffusion record on a 4-class subset of the corpus. The directional asymmetry — software/tech-origin themes arriving in transport and advertising/retail with multi-year lags, the reverse rarely happening — is concrete at the phrase level (Section 3) and reproduces at the theme level under unsupervised LDA topic extraction (Section 5), with within-paradigm robustness under independently-fitted NMF.

Δ KL predicts post-registration §8 mark maintenance in a U-shape at both the filing and token levels, with the token-level gap larger than the filing-level gap and the two should not be conflated. The U-shape is robust to operationalization; the underlying mechanism is not separable with the trademark record alone.

Δ KL is empirically orthogonal-to-negative with patent counts within firm, confirming the construct is distinct from patent-track invention.

The Schumpeter Mark I read of new themes does not survive year-matched baseline contrast, and the discipline of base-rate contrast is what makes the difference.

The paper does not make these claims.

Δ KL does not measure innovation in any strong sense. The defended primitive is lexical resonance on commercial text. The strong-sense “ Δ KL measures innovation” reading is contradicted by the within-firm patent orthogonality and is not licensed by the small-sample validation evidence ($n = 48$ matched firms on BCG/MIT lists, reported in companion construct- validity work but not in this paper).

Δ KL does not identify a return premium on extreme-tail debut filings; an earlier reported +28pp per σ four-year debut excess return is a small-sample right-tail artifact and is dropped.

The LDA/NMF robustness is within the bag-of-words paradigm. A cross-paradigm test — sentence-embedding clustering or graph-based community detection — is the natural methodological extension and is not run here.

The four-class subset cannot exclude class-size or corpus-construction explanations for the software/tech-as-net-exporter asymmetry; a 45-class build with NICE-to-NAICS crosswalk is the natural extension.

What would falsify the diffusion claims. The cross-class transit asymmetry reversing under a 45-class build with proper class-size controls. The theme-level flow asymmetry failing to reproduce under a structurally different topic-extraction method (sentence-embedding cluster or graph community detection). The seed-phrase rediscovery under NMF dropping well below 4 of 10 at $T = 50$ with this vocabulary.

What would falsify the patent orthogonality. The within-firm coefficient reversing under value-weighting or an alternative trademark-owner-to-patent- assignee match.

What would falsify the U-shape. The shape disappearing under joint length-and-volume control on a class-by-class basis, or appearing only in classes where the maintenance-base-rate is also U-shaped in firm characteristics not controlled here.

9 Reproducibility

Code, intermediate data panels, and the integrated working report are available at <https://github.com/ericssilver/novelty>. The full pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command for the four-class corpus build (multi-hour; the underlying

TRTYRAP backfile is approximately 12 GB streamed once). The analysis scripts and their outputs are indexed in the repository `README.md` under “Result lookup.”

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